Progressive Depletion Minting

A Foundation Paper

The Mann Mechanics Framework

An institutional introduction to a non-discretionary supply control system

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Preface

This document introduces a control mechanism called Progressive Depletion Minting and the institutional framework built around it. It is written for senior readers who have not encountered the work before and who need to decide whether it warrants the time of their economists, technical reviewers, and policy advisors.

The paper is deliberately accessible. It does not require a background in monetary economics, control theory, or computer science. Where technical material is necessary, it is introduced and explained on the page. Where the document refers to the underlying mathematics, it points to the technical specifications that support it rather than reproducing them here.

It is also deliberately self-contained in scope. The reader who finishes this paper will not have all the answers about PDM. They will, I hope, have a clear enough understanding of what the mechanism is, what it does, what it does not do, and how the framework is constituted, to decide whether deeper engagement makes sense.

The paper makes one substantive request of the reader: take seriously the possibility that a structural problem in the way bounded resources are governed has now received a mechanical answer. Whether that answer is correct, deployable, and useful in any given context is a question for technical review. Whether it deserves to be reviewed at all is the question this paper exists to answer.

Executive Summary

Every system that manages a bounded resource pool faces the same question: when, and by how much, should the pool be expanded? Money supply, energy capacity, carbon allowances, compute allocation, regulated quotas. The question repeats wherever a finite resource meets unbounded demand.

In almost every existing system, the answer is discretionary. Humans decide. The decision arrives late, the logic is opaque, and the cumulative effect over decades produces the credibility deficits that institutions now face.

Progressive Depletion Minting is a mechanism that replaces the discretionary decision with a deterministic rule. The mechanism contracts continuously, expands only when measurably depleted, and resists expansion progressively as the system grows. The structural properties hold for any valid choice of parameters and any non-negative sequence of inputs. Every operating cycle is recorded in a tamper-evident audit chain.

PDM is domain-agnostic. The same mathematics applies to a central bank, a carbon registry, an energy regulator, a bandwidth allocation system, or a compute pool. What changes from

sector to sector is the meaning of the inputs and the choice of operating parameters. The control logic is identical.

Mann Mechanics is the institutional framework built around the mechanism. It comprises three elements: the mechanism itself, a three-layer licensing model that balances open scrutiny against commercial protection, and an independent certification body called MannCert that is structurally separated from Mann Mechanics to preserve the credibility of conformance assessment.

The work has been timestamped and registered. The patent is pending (UKIPO GB2513172.3, filed August 2025). The reference implementation is open for non-commercial study. The supporting specifications, including the telemetry integrity framework that closes the oracle problem, are published and Safe Creative registered.

This paper is the front door. The technical specifications, the simulator, the licensing framework, and the institutional engagement pathway are all behind it. What follows is a structured introduction to the mechanism, the framework, and the engagement pathway, designed to help senior readers decide whether to step through that door.

Part I: The Problem

1. The Same Failure, Everywhere

Look closely at any major governance failure of the last fifty years. The pattern repeats.

A central bank discovers that the decisions that produced an inflationary cycle are difficult to defend in retrospect, even though each individual decision was defensible at the time. A carbon market's credibility collapses because the mechanism by which credits were issued was opaque enough to permit manipulation. An energy regulator faces a capacity crisis because the system that was meant to manage growth could not respond fast enough to the conditions that warranted expansion. A digital asset protocol fragments because there was no structural constraint on its issuance and competing forks emerged claiming to be the real one.

These look like different problems. They are not. They are the same problem with different vocabulary.

Discretion in the issuance of bounded resources is the structural failure that recurs across sectors, and the credibility cost compounds over decades.

Discretion is slow. The decision arrives later than the condition that warranted it. By the time the committee meets, the executive signs off, or the trader reacts, the underlying state has often moved further. Discretionary expansion lags the system it is meant to govern.

Discretion is opaque. Outside observers cannot verify why a particular decision was made. They can read the minutes, the press release, the explanatory speech. But the underlying logic that mapped state to action is not auditable. Institutions can act in good faith and still face credibility costs because the logic is invisible.

Discretion accumulates. Each individual decision may be defensible in isolation. The cumulative effect over decades is what produces the credibility deficits institutions now face. There is no structural mechanism preventing political incentives from biasing expansion in one direction over time. There rarely is, in any institution operated by humans on multi-decade horizons.

This is not an indictment of the people involved. The committees are usually competent and the executives usually well-intentioned. The problem is structural. It is what happens when a finite resource meets unbounded demand and the only governance available is human judgement.

2. What Existing Approaches Get Wrong

There are three established families of approach to bounded supply control. Each has at least one of the desirable properties. None has all of them.

Approach	Where it fails
Commodity standards (gold, etc.)	Bounded by physical scarcity but unresponsive. The system cannot expand to meet legitimate growth in obligations, producing deflationary crises. The bound is not chosen, it is imposed by geology.
Discretionary fiat	Responsive but unbounded. Expansion decisions are made by humans applying judgement to contested inputs. The decisions arrive late, the logic is opaque, and the cumulative effect over decades produces credibility deficits that no amount of communication can close.
Fixed-schedule digital systems (Bitcoin and derivatives)	Bounded and auditable, but unresponsive in a different way. Issuance follows a pre-committed schedule that cannot adapt to actual obligations. Deflationary by construction, with no mechanism to match supply to real-world demand.

Each of these approaches solves part of the problem. None solves all of it. A system that is bounded but unresponsive produces deflation. A system that is responsive but unbounded produces inflation. A system that is bounded and responsive but undisciplined produces fragmentation.

What is missing is a fourth approach. One that is bounded, responsive, structurally disciplined against runaway expansion, and deterministic in a way that any party can verify. That is the design space PDM occupies.

Part II: The Mechanism

3. PDM in Plain Language

Progressive Depletion Minting is, at its core, a thermostat for bounded resource systems.

A thermostat does not consult a committee about when to switch the heating on. It measures the temperature, compares it to a target, and acts. If the room is too cold, the heater runs. If the room is at temperature, it stops. The logic is transparent enough that anyone can understand it, and there is no discretion in the moment of action. The system designer chose the target temperature; after that, the mechanism takes over.

PDM works the same way. It measures the state of the system, compares the state to a target, and acts. There is no committee deciding whether the action is warranted. The action follows from the measurement, deterministically and visibly, every time.

The mechanism has three components, applied in sequence on every operating cycle.

Component	What it does
Continuous contraction	On every operating cycle, supply is reduced in proportion to measured activity. The contraction is the default state of the system. It runs always, regardless of whether anyone has decided anything. There is no committee meeting and no announcement.
Conditional expansion	New supply is added only when the ratio of supply to obligations falls below a defined threshold. If the ratio is healthy, no expansion occurs. If the ratio is below the threshold, the mechanism calculates a restoration amount. The trigger is a measurement, not a judgement.
Progressive resistance	When expansion does occur, the amount actually added is reduced by a damping factor that depends on how full the system already is. As supply approaches the ceiling, each successive expansion delivers less than the one before. The system becomes structurally harder to inflate the more it has already been inflated.

Together these three components produce a system that contracts continuously, expands only when measurably depleted, and resists expansion progressively as it grows. Discretionary timing and quantity are removed. What remains is a transparent rule, observable to any party with access to the audit chain.

The progressive resistance component is the property that gives the mechanism its name. It is also the most important departure from existing approaches. Commodity standards have hard caps but no graduation in approach. Discretionary fiat has no cap. Fixed-schedule digital

systems have a cap but a pre-committed schedule. PDM has a cap, an adaptive schedule, and an exponentially increasing resistance to expansion as the cap is approached. The system makes its own future expansion harder, by mathematics rather than by judgement.

4. The Five Structural Guarantees

Five properties hold for the state transition function that defines PDM. They hold unconditionally, regardless of who operates the mechanism, what parameters they choose, or what telemetry they feed into it.

- Supply can never go below zero.
- Supply can never exceed the system's hard ceiling.
- Expansion occurs only when the supply-to-obligation ratio falls below the lower stability band.
- Each successive expansion is structurally harder than the one before, because the damping factor decreases monotonically as supply grows.
- Every operating cycle is recorded as a complete trace and chained via SHA-256, producing a tamper-evident audit history that any external party can verify.

The PDM Personal Edition Whitepaper provides the formal specification of the control law, including explicit statements and proofs of the structural invariants of the state transition function. These proofs establish the properties listed above, non-negativity, capacity boundedness, conditional minting, and progressive resistance, directly from the construction of the update function. They do not claim global optimality, full system stability under all possible telemetry conditions, or integrity of input data, which are addressed separately through empirical validation and the telemetry framework. Appendix C of the Whitepaper presents reproducible simulation results verifying these properties against the reference implementation.

Stability beyond these structural guarantees is empirical. The Whitepaper's Appendix C presents seven test regimes empirically verifying all five guarantees, and the Simulator Supporting Documentation extends this with a 625-point parameter stability sweep finding that the contraction coefficient remains positive across all tested configurations. The mechanism returns to its target ratio under noise within the tested parameter range. The results are reproducible by any reviewer running the simulator with the same parameters.

5. Why the Mechanism Generalises

PDM operates on three abstract quantities: supply, obligations, and activity. The control law does not care what these quantities physically represent. The same state transition function applies whether the system being governed is a money supply, an energy grid, a carbon credit registry, a compute pool, or a regulated quota.

This is not aspiration. It is a property of the function itself. The mathematics is identical across sectors. What changes from sector to sector is the mapping from real-world quantities to the three abstract inputs, and the choice of operating parameters appropriate to that sector.

Domain	Supply	Obligations	Activity
Money creation	Money supply	Credit demand	Settlements
Carbon credits	Credits issued	Compliance obligations	Credits retired
Energy grid	MW capacity	Grid demand	Energy delivered
Bandwidth	Capacity	Traffic demand	Throughput
Compute	Compute credits	Job queue	Jobs processed

The cross-sector applicability is what makes PDM a platform-level mechanism rather than a sector-specific innovation. A central bank deploying PDM and an energy regulator deploying PDM are using the same mathematics. The certification process, the audit chain format, and the structural guarantees are common across deployments. Only the parameter choices differ.

The next section walks through three sector applications in detail. They are illustrative, not prescriptive. None has been deployed in production. They are presented to make the abstract mechanism concrete and to give the reader a structured way to evaluate whether a similar architecture might apply to their own context.

Part III: Worked Examples

The following sketches illustrate how PDM could be applied in three different sectors. They are not deployment plans. Each would require detailed sector-specific calibration, regulatory analysis, and institutional design that is properly the work of a Stage 2 Sector Analysis or Stage 3 Implementation Blueprint, not a foundation paper. What the sketches do show is how the same control mechanism takes on a recognisable shape in each context.

6. Carbon Credit Registry

Carbon markets have a structural integrity problem. The credits issued are supposed to represent verifiable emission reductions, retirement commitments, or compliance obligations. In practice, the relationship between credits issued and underlying obligations has been opaque enough to permit episodes of significant manipulation, and confidence in the asset class has suffered accordingly.

A PDM-based carbon registry would replace the discretionary issuance of credits with a deterministic mechanism governed by verified compliance obligations. The architecture would comprise:

- An obligation register of verified compliance liabilities: regulated emitters' reported emissions, retirement commitments, and compliance deadlines, each entered subject to dual-key validation by independent verification bodies.
- A fixed ontology restricting admissible obligations to regulated compliance instruments with identified liable parties and statutory due dates. Speculative credits, voluntary commitments without enforceable counterparties, and aspirational targets are excluded by construction.
- A control law instance governing the issuance of new credits against a hard cap, with progressive resistance applied as the cap is approached. The cap itself is a policy choice; the mechanism enforces it without exception.
- Hash-chained publication of both the obligation register state and the control law transitions, enabling independent market participants to verify that issuance is responsive only to authentic compliance demand.

The structural property a participant in this market would have, that they do not have today, is verifiability. They could check, at any time, that every credit in circulation corresponds to an obligation that was validated by independent parties, has not yet expired, and was not issued in excess of the system's progressive resistance schedule. The credibility of the credit would no longer depend on trust in the issuer. It would depend on inspection of the chain.

Sector-specific performance claims for this architecture are unproven in the absence of real-world deployment data. The sketch is presented as a candidate design, not as a validated production system.

7. Central Bank Monetary Issuance

Central banks face the credibility deficit described in Part I more acutely than almost any other institutional category. Each individual policy decision is defensible. The cumulative trajectory over decades produces the structural critique that no individual decision-maker can fully address through communication.

A PDM-based monetary deployment would not replace the central bank. It would change the locus of decision from the moment of issuance to the moment of parameterisation. The architecture would comprise:

- A statutorily defined ontology of admissible monetary obligations: unpaid invoices, payroll obligations, tax assessments, settlement backlogs, contractual deliveries. The ontology is set in law, not by executive action.
- An independent obligation register operated by a chartered institution, with dual-key validation by structurally separate authorities such as a national audit office, a central clearing counterparty, and a tax administration operating under judicial review.
- A control law instance operating PDM's burn, conditional mint, and progressive damping functions against the aggregate obligations produced by the register.
- A constrained override authority exercisable only by named officers under statutory conditions, with mandatory public declaration, defined deviation bounds, time-limited effect, and full logging in the audit chain.

The central bank retains authority over the parameters: the system's hard ceiling, the stability band, the contraction coefficient. What it relinquishes is discretionary timing and quantity of issuance once the parameters are set. It also gains a structural defence against the accusation that issuance is opaque, because every cycle is observable to any party with access to the audit chain.

Sovereignty is preserved at the level that matters. The institution chooses the rules; the mechanism executes them.

This is not a small claim, and it is not one that should be evaluated quickly. A monetary deployment of PDM would require constitutional or treaty-level commitment to the ontology, a structural separation between fiscal authority and obligation validators that is real rather than nominal, and a legal framework that recognises cryptographic attestations as admissible evidence of obligation. These are non-trivial preconditions, and they are explicitly acknowledged in the supporting specifications. The framework is designed for jurisdictions willing to operate under rule of law. It will not protect against a state that refuses to be bound by its own rules.

8. Energy Grid Capacity

Energy grids face a different version of the same problem. Capacity must be matched to demand, but demand is variable, capacity is expensive to add, and the consequences of getting the balance wrong include both blackouts and stranded assets. Most existing capacity markets rely on discretionary signalling by the regulator or operator, who decides when to encourage new capacity and when to allow existing capacity to retire.

A PDM-based capacity mechanism would replace those discretionary signals with measured ones. The architecture would comprise:

- An obligation register of contracted load, reserve margin commitments, and interconnection obligations, each validated by the relevant grid operator and an independent system reliability body.
- A control law instance operating against a hard ceiling defined by physical interconnection capacity, with progressive resistance applied as that ceiling is approached.
- Continuous contraction reflecting actual energy delivered, so that the system tightens as utilisation rises and relaxes as utilisation falls.
- A publication layer making both the obligation register and the control law transitions observable to all market participants, so that capacity investment decisions are made against verifiable rather than asserted system state.

The property such a system would have is dynamic discipline. Capacity expansion would track measured obligations rather than projected demand. As the grid approached its physical ceiling, the mechanism's progressive resistance would naturally favour demand-side response, storage, and efficiency over additional generation, without anyone having to decide that this was the policy. The policy would be implicit in the mathematics.

Performance claims for this architecture are again unproven without deployment data. The sketch is one candidate among others. What the sketch shows is that the same mechanism that governs monetary issuance can govern grid capacity, with a different set of inputs and different operating parameters but the same control logic.

Part IV: The Telemetry Layer

9. The Oracle Problem

The structural guarantees of the control law hold given valid telemetry. They make no claims about the authenticity of the inputs themselves. If the obligations and activity values fed into the mechanism are manipulated, the mechanism will produce expansion in response to false signals. The control law is verified; the inputs to it are a separate engineering problem.

This is not a flaw. It is a deliberate separation of concerns. A control law and an oracle are distinct engineering problems. Conflating them produces systems that are either mathematically elegant but operationally fragile, or operationally robust but mathematically opaque. Separating them makes both tractable.

The separation creates a clear attack surface. If the obligation register can be manipulated, then even a formally verified control law will expand supply in response to false signals. Three failure modes deserve explicit treatment: discretionary redefinition of what counts as an obligation, unilateral insertion of synthetic entries by a single actor, and temporal manipulation in which obligations never expire and accumulate indefinitely.

10. The Telemetry Integrity Framework

The companion specification, titled O_n as Auditable Ledger, addresses these failure modes explicitly. It defines:

- A fixed ontology of admissible obligation types, set by statute or equivalent instrument, that cannot be redefined by executive action.
- A dual-key validation requirement on every entry, with each obligation confirmed by at least two structurally independent sources before admission.
- Temporal expiry rules that automatically remove unsettled obligations after a defined period, preventing the accumulation of stale claims.
- Netting and deduplication procedures that remove circular and duplicate obligations before they enter the aggregate.
- Cryptographic attestation and hash-linked reporting, distinct from but complementary to the control law's own audit chain.
- Structural separation of powers between the authority that benefits from issuance and the authority that validates the obligations driving it.
- A constrained override protocol for genuine emergencies, requiring named declaration, bounded deviation, time-limited effect, and full audit logging.

The combined effect of the control law and the telemetry framework is that manipulation of the system cannot be absorbed into the issuance rule. Any deviation from the rule must occur as an explicit, named, time-limited override that is cryptographically logged and publicly verifiable.

The mechanism does not eliminate the possibility of override. It eliminates the possibility of hidden override.

This is the structural improvement. Every previous monetary, capacity, or compliance system has permitted manipulation to be absorbed into the issuance rule. The PDM control law combined with the telemetry framework forces manipulation into the open. Capture that occurs through openly declared, repeated override remains possible. Capture that relies on hidden intervention is foreclosed.

Part V: The Institutional Framework

11. Mann Mechanics and MannCert

The mechanism alone is not enough. A patented mechanism with no governance model fragments into unauthorised forks. A proprietary black box with no specification cannot be trusted by the institutions that would adopt it. The institutional framework is what determines whether the mechanism becomes deployable infrastructure or remains a paper.

Mann Mechanics is the institutional layer. It comprises two structurally separated entities.

Mann Mechanics	MannCert
Framework development and stewardship. Institutional engagement and advisory. Commercial licensing. Sector analysis and implementation planning.	Independent conformity assessment. Standards definition and maintenance. Public registry of certified deployments. Enforcement, appeals, and certification withdrawal.

This separation is by design. It ensures that conformity assessments are conducted independently of commercial licensing relationships. MannCert's certification decisions are not subject to approval by Mann Mechanics. A deployment that fails MannCert assessment does not lose its commercial licence automatically, but it cannot continue to display the conformance mark. The patent holder cannot grant or withdraw certification as a commercial favour.

This is the governance feature most likely to determine whether the framework is taken seriously by sophisticated institutional reviewers. A reviewer evaluating a patented mechanism with associated certification will look first at whether the certifier is independent of the patent holder. If certification is a commercial function of the rights holder, the certification is worthless. If it is structurally separate, the certification has the credibility of a third-party audit.

12. The Three-Layer Licensing Model

The licensing structure is designed to balance three objectives that exist in tension. Open scrutiny is required for academic credibility and independent verification. Commercial protection is required to fund continued development and to prevent fragmentation. Quality assurance is required for institutional adoption where public trust depends on correct implementation. The three-layer model addresses all three.

Layer	Who it is for	Rights	Obligations
Open Study	Researchers, academics, non-	Read, study, replicate, simulate, publish analysis	Attribution required; no commercial deployment

Layer	Who it is for	Rights	Obligations
	commercial implementers		
Commercial Licence	Companies deploying PDM in production for commercial gain	Deploy a PDM-based mechanism in a named commercial context	Compliance with licence terms; periodic reporting on deployment
Institutional Certification	Operators of regulated, public-trust, or mission-critical deployments	Commercial rights plus right to display the MannCert conformance mark	All Layer 2 obligations plus conformance audit and ongoing review

Layer 1, Open Study, is what makes academic and independent scrutiny possible. The reference simulator, the Personal Edition Whitepaper, the Simulator Supporting Documentation, and all companion papers are published openly. A researcher can replicate the mechanism, run their own simulations, publish their own analysis, and propose refinements through the public comment process. They cannot deploy the mechanism commercially without entering Layer 2, but they can study it freely.

Layer 2, Commercial Licence, is the operating layer for sectors that wish to deploy PDM in production. Licence terms are agreed on a per-deployment basis. Parameters considered in fee-setting include sector, scale, geographic scope, and the commercial model of the licensee's deployment. The licence does not confer the right to display the MannCert conformance mark; that is reserved for Layer 3.

Layer 3, Institutional Certification, is required for deployments where public trust, regulatory standing, or mission-critical function depends on verifiable correct implementation. Certification is conducted by MannCert, is subject to periodic audit and renewal, and may be withdrawn for non-conformance. A certified deployment is entitled to display the MannCert conformance mark and is listed in the public registry of certified deployments.

The full structure, including licence scope, fee model, governance, and revocation conditions, is specified in the PDM Licensing and Certification Framework document. It is published, Safe Creative registered, and available for review.

13. The Engagement Pathway

Institutional engagement with Mann Mechanics proceeds through a structured pathway. Each stage is self-contained. Institutions may pause, defer, or discontinue engagement at any point without obligation. The pathway is designed to allow institutions to engage at the depth appropriate to their stage of evaluation.

- Stage 1, Institutional Briefing. A structured written advisory engagement responding to up to ten institutional questions. The output is a written advisory document analysing the framework in the context of the requesting institution's domain. Suitable for senior leadership teams forming an initial view.
- Stage 2, Sector Analysis. A detailed analytical report examining how PDM could operate within a defined sector or institutional environment. The report evaluates positioning, parameterisation, governance implications, and policy relevance, providing a basis for internal institutional review.
- Stage 3, Implementation Blueprint. A tailored architectural document describing how the framework could be structured within the operational environment of the commissioning institution. Supports internal planning and governance evaluation. Does not constitute operational implementation.

Detailed terms, including fees, payment structures, and confidentiality provisions, are set out in the Mann Mechanics Institutional Engagement Fee Schedule. Engagements are conducted through structured written communication to ensure precision of interpretation and an auditable record of advisory work.

Part VI: Founder's Note

14. How This Work Came About

Institutional readers are entitled to know who built this and why they should be taken seriously. This section is the answer to that question. It is short by design. The framework's credibility should rest on the framework, and the technical and licensing specifications are designed to be evaluated on their merits regardless of the author.

My name is Valraj Singh Mann. I was born in Greater London in 1977 and have lived in the United Kingdom my entire life. I am the founder of Mann Mechanics Ltd and the inventor of Progressive Depletion Minting. The patent application was filed at the UK Intellectual Property Office in August 2025 under reference GB2513172.3. Progressive Depletion Minting, Mann Mechanics, MannCert and MannMark are all UK registered trademarks (UK00004244285, UK00004237915, UK00004326618, and UK00004326620 respectively).

The work originated, as is often the case with control mechanisms, in a different domain than the one that ultimately gave it shape. It began in mid-2025 as a tokenomics design problem: how to create a charity digital asset, intended to support feeding the homeless, whose supply curve responded to actual usage rather than a pre-committed schedule. The mathematics that resolved that problem turned out to be domain-agnostic. The same control law that disciplined a charity token's issuance could discipline a money supply, a carbon registry, or any other bounded resource pool. Once that generalisation became clear, the framework reorganised itself around it.

My background is unconventional for this kind of work, and I think that has been an asset. Thirteen years in enterprise IT operations management gave me a working understanding of how systems fail in practice, which is rarely how they fail in theory. Cinematography and UK crypto media production work, including production at NFT.London and engagement with parliamentary all-party groups on AI, blockchain, and the metaverse, gave me exposure to how technology meets policy at the level where decisions are actually made. Long-form editorial work on Islamic finance and capital markets gave me a vocabulary for thinking about supply discipline that does not depend on any particular monetary tradition. Building and scaling a national retail distribution business through Sweet Stuff Ltd gave me the discipline of managing real cash flow against real obligations, which is not the same thing as modelling them.

None of these are the standard credentials for proposing a control mechanism for monetary or capacity systems. I think that is actually the point. The structural failure mode I described in Part I is not a problem that the inhabitants of any single field have solved, in part because the failure spans fields. The problem is governance of bounded resources under unbounded demand, and it shows up the same way in money, in energy, in carbon, and in compute. Recognising it required moving across enough domains to see the pattern. Formalising it required spending the time to write down the mathematics that disciplines it.

That is what Mann Mechanics is. It is the institutional vehicle for the framework that came out of that work. Its function is to steward the specification, to engage with institutions on terms appropriate to their stage of evaluation, and to maintain the structural separation from MannCert that preserves the credibility of conformance assessment. It is not designed to grow rapidly or to capture the value of the patent for its own sake. It is designed to be the institutional layer that allows the framework to be evaluated, deployed, and stewarded over the long term.

There is one thing I want to say directly. This work was undertaken in the belief that the credibility deficits institutions face are real, that they have a structural cause, and that the structural cause is addressable through mechanism design. I do not believe the framework solves every problem. I do believe it solves one, and that one is foundational to most others. The case for engagement is the case that this is worth verifying.

15. What Mann Mechanics Is Not

Honest demarcation is part of credibility. The following are explicitly out of scope for Mann Mechanics.

- Mann Mechanics is not a consultancy in the conventional sense. The engagement pathway is designed to support institutional evaluation of a specific mechanism, not to provide general governance or strategy advice.
- Mann Mechanics is not a financial services firm. It does not give investment, trading, regulatory, or financial advice, and the framework does not constitute any such advice.
- Mann Mechanics is not a vendor of operational systems. The framework specifies a mechanism and a governance structure. Operational deployment requires sector-specific calibration, regulatory analysis, and engineering work that is properly done by deploying institutions, possibly with the support of a Stage 3 Implementation Blueprint.
- Mann Mechanics is not a political project. The framework is structurally neutral. It governs the procedure of expansion, not the substance of policy. Its adoption does not commit an institution to any particular policy direction.
- Mann Mechanics is not a substitute for sovereign authority. Institutions retain full authority over the parameters that govern any deployment. The framework's structural guarantees apply within the parameters institutions choose. It does not prescribe what those parameters should be.

Part VII: What Is Being Asked

16. Of the Reader

This paper is the first document in the engagement pathway. It does not request adoption, commitment, or operational change. It asks the reader to consider one question: does the work warrant the time of the people in your institution who are equipped to evaluate it?

The people who are equipped to evaluate it will vary by institution. In a central bank, it might be the chief economist, the head of monetary operations, and a senior policy advisor. In an environmental regulator, it might be the head of compliance markets and an economist with carbon market experience. In an energy regulator, it might be the head of capacity mechanisms. In a sovereign fund or development bank, it might be a strategy unit charged with evaluating governance innovations.

If, having read this paper, you can identify the people in your institution who would be the right reviewers, the request is to put the framework in front of them. That is the only ask at this stage. No commitment, no resource allocation, no public position. Just the decision to allow technical review.

If a Stage 1 Briefing is the appropriate format for that review, the engagement pathway provides for it. If it is not, the open materials are sufficient for unstructured evaluation, and the cost to your institution is only the time of your reviewers.

17. Of the Framework

It is reasonable for the reader to ask what the framework is asking of itself. The answer is the discipline that the rest of this paper describes. Mann Mechanics commits to operating the framework according to the published specifications. MannCert commits to operating its certification function independently of Mann Mechanics. The reference materials remain open for non-commercial study. Commercial licence terms remain published and amendments to them are announced with reasonable notice.

This is a structural commitment, not a rhetorical one. The framework's credibility depends on the framework operating as specified. Departures from the published specifications would be visible to any party tracking the work, and the work is published precisely so that they are visible. The framework holds itself accountable to the same kind of structural transparency it asks of its deployments.

Closing

The case for taking PDM seriously is that the structural failure described in Part I is real, recurring, and not currently addressed by any existing approach to bounded supply control. The case for taking Mann Mechanics seriously is that the institutional framework around the mechanism is structurally sound, transparently published, and designed for the long term.

Neither case is closed by this paper. They are opened by it. The technical specifications, the licensing framework, the simulator, and the engagement pathway are the means by which serious readers can verify the cases for themselves. Whether they do is the decision this paper exists to make easier.

The framework provides the mechanism, the specifications, the certification structure, and the engagement pathway. What it cannot provide is the institutional decision to engage.

That decision belongs to the reader. This paper has tried to give them the structured ground on which to make it.

Status and Disclaimer

Progressive Depletion Minting is covered by a pending UK patent (UKIPO GB2513172.3, filed August 2025) held by Mann Mechanics Ltd. The patent is at pending status; grant has not yet been issued. The marks Progressive Depletion Minting, Mann Mechanics, MannCert and MannMark are UK registered trademarks (UK00004244285, UK00004237915, UK00004326618, and UK00004326620 respectively), all in the name of Valraj Mann.

The Personal Edition Whitepaper, Simulator Supporting Documentation, Telemetry Companion Paper, Licensing and Certification Framework, Institutional Brief, Technical Overview, and Equilibrium Demonstrator Companion are published on MannMechanics.com. The Whitepaper additionally has a canonical version in the public GitHub repository at github.com/mannmechanics/pdm-personal-edition-reference and is registered with Safe Creative under Work ID 2601084210286 (8 January 2026). The Telemetry Companion Paper and the Licensing and Certification Framework are also Safe Creative registered (references 2604245385991 and 2604245386219 respectively, timestamped 24 April 2026).

This Foundation Paper is provided for institutional reference and does not constitute legal, financial, regulatory, or monetary policy advice. It is not an offer to license, a grant of rights, or a binding commitment. Binding terms are set out in individual licence and engagement

agreements executed in writing. Sector-specific performance claims are unproven in the absence of real-world deployment data and are flagged as such throughout the specifications.

Mann Mechanics Ltd and MannCert Ltd are separately constituted entities with distinct functions. Their structural separation is a deliberate governance feature and should not be conflated.

Institutions remain solely responsible for their own evaluation, decision-making, and policy outcomes.

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